

Q6 — Student Table with FK – Count Students Per Course using GROUP BY*Note: Course column in Student references CourseDetails to enforce referential integrity.*

```

CREATE TABLE CourseDetails (
    CourseID    NUMBER(5)    PRIMARY KEY,
    CourseName  VARCHAR2(100) NOT NULL
);

INSERT INTO CourseDetails VALUES (1, 'CSE-AI');
INSERT INTO CourseDetails VALUES (2, 'CSE');
INSERT INTO CourseDetails VALUES (3, 'IT');

COMMIT;

CREATE TABLE Student (
    StudentID   NUMBER(5)    PRIMARY KEY,
    Name        VARCHAR2(50) NOT NULL,
    CourseID    NUMBER(5),
    CONSTRAINT fk_course FOREIGN KEY (CourseID)
        REFERENCES CourseDetails(CourseID)
);

INSERT INTO Student VALUES (1, 'Rahul', 1);
INSERT INTO Student VALUES (2, 'Priya', 1);
INSERT INTO Student VALUES (3, 'Amit', 2);
INSERT INTO Student VALUES (4, 'Sneha', 3);
INSERT INTO Student VALUES (5, 'Kiran', 1);

COMMIT;

SELECT cd.CourseName, COUNT(s.StudentID) AS Student_Count
FROM CourseDetails cd
LEFT JOIN Student s ON cd.CourseID = s.CourseID
GROUP BY cd.CourseName
ORDER BY cd.CourseName;

```

Expected Output:

Table created.
1 row created. (x3)
Commit complete.

Table created.
1 row created. (x5)
Commit complete.

COURSENAME	STUDENT_COUNT
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CSE	1
CSE-AI	3
IT	1

3 rows selected.